A divider with gain

Evolution with signed values finds negative resistance; its power is booked as model-supplied — CLAUDE.md §3c, §4

kronspice simulation report · 2026-11-07 · seed 5, population 30 × 30 generations

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Summary

A two-resistor divider is asked for a voltage gain of 2 (6.02 dB) at 1 kHz. No passive divider can do it: the only solution is a lower leg of −2 kΩ, so the search has to cross zero and settle on a negative value. The example, from milestone M5 (value genome), shows the *sign: either* gene and the energy ledger's model-supplied column.

- **Result:** after 30 generations (842 evaluations, seed 5) R2 = −2000.86 Ω, a gain of 6.017 dB (error 1.9×10^{-4}). The exact answer is R2 = −2 kΩ.
- **The negative resistor delivers power:** $v^2/|R| = 2^2/2000 = 1$ mW; the run reports model-supplied power 0.000999 W.
- **It is never booked as inflow from the exterior.** An ideal negative resistance stands for a tunnel diode, Gunn diode or op-amp negative-impedance converter whose bias supply the model leaves out.
- **The template circuit** (both legs 1 kΩ, gain ½) gives the same answer by MNA and by Kron's torn solve, with power invariant under tearing.

Why gain needs a negative leg

The divider's transfer function is H = R2/(R1 + R2). With R1 = 1 kΩ and both resistors positive, H is always between 0 and 1. Setting H = 2 gives R2 = −2 R1 = −2000 Ω: H = −2000/(1000 − 2000) = 2. So the target is reachable only if the sign of R2 can change.

CLAUDE.md §3c separates two transformations of a resistor. *Duality* (exchanging voltage and current, as a gyrator does) maps R to 1/R, still positive. *Sign inversion* (time reversal, or a negative-impedance converter) maps R to −R: dissipation becomes generation. A negative resistance is accepted in netlists and genomes, as in ngspice, and delivers $v^2/|R|$. Its power is **model-supplied** — listed per element, balancing the ledger, never counted as inflow — and its thermal port is not modeled.

Circuit, gene and objective

```
V1 in  0   DC 1 AC 1
R1 in  out 1k
R2 out 0   1k          gene: 10 Ω … 100 kΩ in magnitude, sign either
.ac lin 1 1k 1k
```

objective	target	error
gain of V(out)/V(in) at 1 kHz	6.0206 dB (×2)	|gain − 6.0206| in dB / 20

With *sign: either* the sign is its own gene, flipped by mutation, so the log-scale magnitude and the sign evolve separately. GA settings (config.json): population 30, 30 generations, seed 5, mutation step σ = 0.3 decades, target fitness 10^{-4} (not reached, so the run goes the full 30 generations).

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Result

quantity	evolved (seed 5)	exact
R2	−2000.86 Ω	−2000 Ω
gain at 1 kHz	6.017 dB	6.0206 dB
fitness error	1.9×10^{-4}	0
evaluations	842	
model-supplied power	0.000999 W	$2^2/2000$ = 1 mW

The run prints:

```
model-supplied power 0.000999 W (idealized active elements: a real circuit needs a supply for it)
```

The evolved best.cir opens in ngspice, LTspice and QSpice as it is (ngspice accepts negative resistors too).

Ledger and twin check

- The negative resistor's power is booked in the ledger's model-supplied column, per element. A fitness that measures harvested energy must count this column as an input; *evo.ModelSuppliedLimit* is the objective that bounds it.
- Template circuit (both legs 1 kΩ) at 1000 Hz, from the export results sheet: V(in) = 1 at phase 0.00°, V(out) = 0.5 at phase 0.00° by both MNA and Kron's torn solve.
- Kron's solve tears it automatically (ground as common bus, Happ Ch. XII §4) into 2 pieces of [2 1] unknowns joined by 1 link (R1, current 0.0005 at phase 0.00° A, voltage 0.5 at phase 0.00° V); the interconnection system has order 1 against 3 unknowns for the whole circuit.
- Torn against full solve: largest node-voltage difference 0; power V·I* taken by the R, L, C branches 0.0005 at phase 0.00° (torn) against 0.0005 at phase 0.00° (untorn) — power is invariant under tearing (Roth 1959 §5).

Limits

- The negative resistor is ideal: it has no bias supply, no voltage or current range and no frequency limit. A real tunnel diode, Gunn diode or negative-impedance converter takes the 1 mW from its supply.
- One frequency point (1 kHz); the circuit has no reactive elements, so the gain does not depend on frequency.

Reproducing the results

```
go run ./cmd/evolve -config examples/negative_resistance_gain/config.json   # about 2 s
go run ./cmd/kronspice runs/<run>/best.cir                                   # re-simulate the result
KRONSPICE_EXPORT=1 go test ./internal/export                                 # export/ sheets
```

See also doc/M5.md, "Negative resistance and nonlinear elements", and the other config-driven M5 examples rc_lowpass_values and rectifier_filter.

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